Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

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Dataset Overview

3,900

Purchase Records

Transactional data analyzed

18

Data Columns

Customer and purchase details

37

Missing Values

Review ratings imputed

Key Features

- Customer demographics
- Purchase details
- Shopping behavior

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Data Preparation Process

01

Data Loading

Imported dataset using pandas

02

Initial Exploration

Checked structure and summary statistics

03

Missing Data Handling

Imputed missing values using median ratings

04

Column Standardization

Rename to snake case for readability

05

Feature Engineering

Created age groups and purchase frequency

06

Database Integration

Loaded into PostgreSQL for SQL analysis

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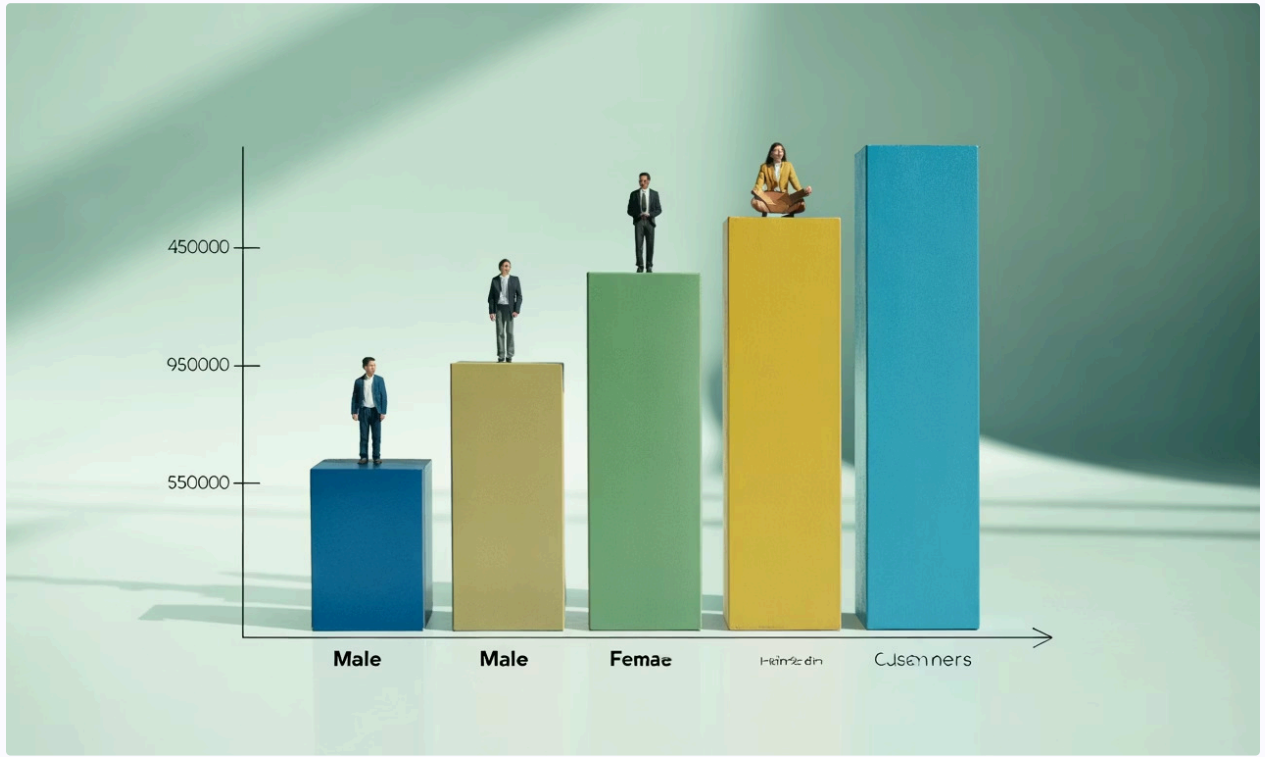
Revenue by Gender



Key Insight

Revenue comparison between male and female customer segments

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High-Spending Discount Users

Strategy

Identified customers using discounts but spending above average

Insight

Discounts drive volume without sacrificing high-value purchases

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Top Products by Rating

Analysis

Products with highest average review ratings identified

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Shipping Type Comparison

Standard Shipping

Average purchase amounts

Express Shipping

Higher spending customers

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Subscribers vs. Non-Subscribers

Subscribers

Average spend and total revenue comparison

Non-Subscribers

Revenue and spending patterns analyzed

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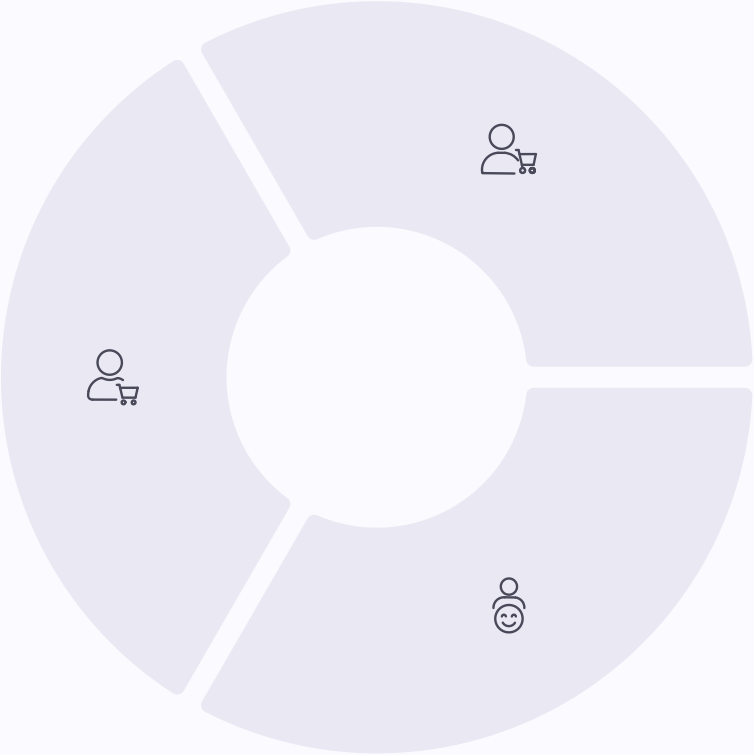
Customer Segmentation



Returning

Multiple purchases

New

First-time buyers

Loyal

Consistent shoppers

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Business Recommendations

1

Boost Subscriptions

Promote exclusive benefits for subscribers

2

Customer Loyalty Programs

Reward repeat buyers to move them into "Loyal" segment

3

Review Discount Policy

Balance sales boosts with margin control

4

Product Positioning

Highlight top-rated and best-selling products in campaigns

5

Targeted Marketing

Focus on high-revenue age groups and express-shipping users

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